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Department of Computer Science and Engineering Mini Project (BCS586) Review – 0

Indian Market Portfolio Intelligence & Backtesting Platform

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Overview

Problem

- Financial markets are highly volatile
- Investors often adopt strategies without risk evaluation
- Existing tools focus mainly on price charts, not strategy robustness

Proposed Solution

Machine learning based **Portfolio Intelligence & Backtesting Platform**

Capabilities

- Strategy Backtesting
- Market Regime Detection
- Risk & Performance Analysis
- ML-based Strategy Recommendation

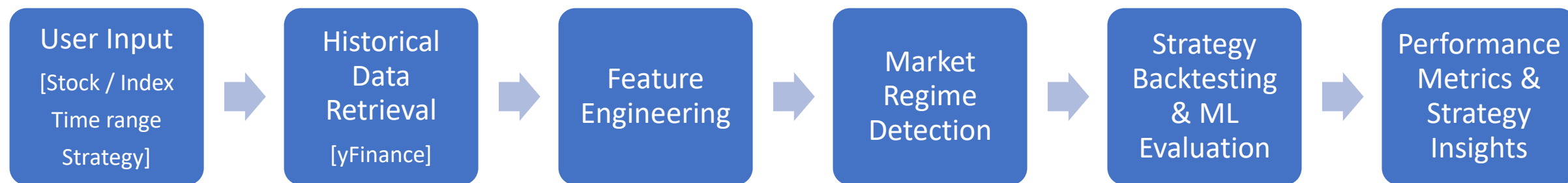
Goals

- Develop a **machine learning–augmented platform** for evaluating trading strategies using historical market data.
- Provide **data-driven insights** into strategy performance and investment risk.
- Enable investors and learners to **analyze trading strategies** under different market conditions.

Objectives

1. Develop a structured backtesting engine for predefined trading strategies.
2. Detect market regimes using unsupervised learning techniques.
3. Evaluate strategy performance across different market conditions.
4. Predict strategy suitability using supervised ML models.
5. Estimate downside risk and volatility using forecasting models.

System Workflow



Strategy Library

Strategy	Purpose
Buy & Hold	Long-term benchmark strategy
Moving Average Crossover	Trend-following strategy
RSI Strategy	Detects overbought / oversold conditions
Momentum Strategy	Invests in stocks with strong recent returns
Mean Reversion	Assumes price returns to historical mean
Auto-Recommend Mode	Strategy based on current regime

Applications

1. Investment Decision Support

Helps investors identify suitable trading strategies.

2. Financial Research

Enables researchers to study market regimes and strategy performance.

3. Educational Tool

Useful for students learning financial analytics and machine learning in finance.

4. Risk Analysis

Provides insights into drawdowns, volatility, and downside exposure.

5. Portfolio Analysis

Provides insights into multi-stock portfolios by analyzing risk-adjusted returns and drawdown metrics.

Requirement Specifications

Minimum Hardware Requirements:

- **Processor:** A modern multi-core processor
- **RAM:** 8GB
- **Storage:** 5 GB of space for the project, dataset, and libraries.
- Internet connectivity for data retrieval

Minimum Software Requirements:

- **Operating System:** Windows 10/11, macOS, or a Linux distribution
- **Programming Language:** Python 3.10+
- **Libraries:** FastAPI, Scikit-learn, Pandas, NumPy, SciPy, Plotly, yfinance / NSEpy
- **Database:** PostgreSQL
- **Development Environment:** Any standard IDE like VS Code or PyCharm.

Data Set

Data Source

- Historical stock market data from Yahoo Finance via the yfinance API.

Training Dataset

- Historical data from **NIFTY-50 companies**
 - ~50 stocks, 20+ years of history (≈ 2000 – 2023)
 - ~5000+ trading days per stock

Dataset Fields

- Date | Open | High | Low | Close | Volume (OHLCV)

Runtime Data Retrieval

- Daily data fetched dynamically based on user-selected stock and time range.

Derived Features for ML

- Returns, Volatility, Momentum, Maximum Drawdown

Total Scale

- $\approx 250,000$ + historical data points used for training and analysis.

References

Journal Papers:

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- [2] L. Yu, Y. Chen, and Y. Zheng, “A Multi-Market Data-Driven Stock Price Prediction System: Model Optimization and Empirical Study,” IEEE Access, vol. 13, pp. 65884–65899, 2025.
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Website:

- [4] [Yahoo Finance – Historical Stock Market Data](#) accessed on 11-03-2026